

An explicit Collatz block-growth bound below $\log_2(3)$

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Private research draft - independent mathematical review pending

A specialization of Bugeaud's p-adic interpolation theorem gives the candidate all-orbit block-growth bound below, with base $\lambda = \log_2(3) - 1/93$, approximately 1.574210. The argument combines a fourth-power normalization, two explicit interpolation ranges, a dyadic tail, and a bounded restart construction. Published cycle-size estimates then give $K < 15.17m \cdot \lambda^m$ for hypothetical cycles with $m \geq 515620$ local minima. The complete derivation is presented here. Independent mathematical review and priority verification remain pending. The Collatz conjecture remains open.

Statement and conventions

Use $T(n) = (3n+1)/2$ for odd positive n and $T(n) = n/2$ for even n . A block is a maximal odd run followed by a maximal even run. At the odd start n_t define $k_t = v_2(n_t + 1)$ and $y_t = \log_2(n_t + 1)$. Put $\delta = \log_2(3)$, $\beta = \delta - 1$, and

$$\varepsilon = 1/93, \lambda = \delta - \varepsilon, \\ C = 92, B = 2^{15}, A = (\delta/\lambda)^{28}.$$

The candidate theorem, for every positive orbit, is

$$k_t \leq A \cdot y_0 \cdot \lambda^t \quad (t \geq 0), \\ y_t \leq \delta \cdot A \cdot y_0 \cdot \lambda^{t-1} \quad (t \geq 1).$$

The index counts blocks and $\lambda > 1$, so these bounds allow unbounded trajectories. For a hypothetical nontrivial cycle, m counts local minima and K is the sum of the m odd-run lengths. The cycle consequences are

$$K < 1.51m \cdot \lambda^m, 1024 \leq m \leq 515619, \\ K < 15.17m \cdot \lambda^m, m \geq 515620.$$

The proposed contribution is the explicit specialization and its global application. P-adic interpolation and normalized-exponent refinements are established methods; novelty of this application is unconfirmed.

Exact block identities

Write $n = a \cdot 2^k - 1$, with a a positive odd integer. Let $\ell \geq 1$ be the following maximal even-run length, $b_0 = 2^\ell - 1$, and n' the next odd block start. Then

$$n' = (a \cdot 3^k - 1)/2^\ell, \\ k' = v_2(a \cdot 3^k + b_0) - \ell, \\ y = k + \log_2(a).$$

With $D = \ell + \beta \cdot \log_2(a) - 1 \geq 0$, elementary logarithmic estimates give

$$y' \leq \delta \cdot y - D, \\ y'' \leq \delta \cdot y - D + \beta \cdot k'. \quad (1)$$

For example, $n' + 1 < (a \cdot 3^k)/2^\ell + 1 < 2 \cdot a \cdot 3^k/2^\ell$ gives the first bound. The second also uses $y'' \leq y' + \beta \cdot k'$, which follows by applying the first bound to the next block and discarding $\ell' - 1 \geq 0$. Always $k \leq y$ and $y' \leq \delta \cdot y$.

The published interpolation criterion

We use Bugeaud (1999), Theorem 1, rational clause (6) [1]. In the rational, 2-adic setting used below, the theorem has the following form. For two rational 2-adic units α_1, α_2 and positive integers $b_1=H, b_2=1$, let $\Lambda=\alpha_1^H-\alpha_2$ be nonzero. Our choices have $g=1, u=0$, field parameters $D_{\text{field}}=e=f=1$ and $t=1$, with $v_2(\alpha_1-1) \geq E > 1$ and $v_2(\alpha_2-1) \geq 2$.

Rename the theorem's integer K as M . For integers $M \geq 3, L \geq 2$ and positive integers R_1, S_1, R_2, S_2 , set

$$\begin{aligned} R &= R_1 + R_2 - 1, S = S_1 + S_2 - 1, N = ML, \\ \gamma &= 1/2 - N/(6RS), P_M = \text{product}(i!, i=1, \dots, M-1), \\ b &= [R-1+(S-1)H] \cdot P_M^{-2/[M(M-1)]/2}. \end{aligned}$$

The two cardinality requirements will be checked explicitly in each range. If they hold and

$$\begin{aligned} M(L-1) \cdot E \cdot \ln 2 - 3 \ln N - (M-1) \ln b \\ - \gamma \cdot L \cdot R \cdot h(\alpha_1) - \gamma \cdot L \cdot S \cdot h(\alpha_2) > 0, \end{aligned}$$

then $v_2(\Lambda) \leq E(ML-1/2) < EML$. For a reduced rational p/q , its logarithmic height is $h(p/q) = \ln(\max(|p|, |q|))$. Negative rational arguments are allowed by the theorem. Once the stated cardinalities hold, this criterion requires no multiplicative-independence assumption.

Factorial normalization and numerical logarithms

For every integer $M \geq 800$,

$$P_M^{-2/[M(M-1)]} < (23/5)/(M-1). \quad (2)$$

Indeed $\ln(i!) \geq i \ln(i) - i + 1$ by integrating $\ln x$. Since $x \ln x$ is increasing for $x \geq 1$, a second integral comparison gives

$$\begin{aligned} \ln P_M &\geq ((M-1)^2/2) \ln(M-1) - (M-1)^2/4 + 1/4 \\ &\quad - M(M-1)/2 + M-1. \end{aligned}$$

Multiplying by $-2/[M(M-1)]$ gives a quantity at most $-\ln(M-1) + 3/2 + \ln(M-1)/M$. For $M=800u, u \geq 1$, $\ln(M-1) < \ln 800 + \ln u < 6.7 + u - 1 < 7.2u$, using $\ln u \leq u - 1$. Consequently $\ln(M-1)/M < .009$. Finally $1.509 < \ln(23/5)$, proving (2). We may also use the weaker factor $5/(M-1)$.

The numerical bounds used throughout are

$$\begin{aligned} .693 < \ln 2 < .6932, \ln 3 < 1.099, \ln 264 < 5.576, \\ \ln(7/4) < .56, \ln 800 < 6.7, \ln 11 < 2.4. \end{aligned}$$

The additional bounds $\ln 90 < 4.5$ and $\ln 150 < 5.011$ follow from $\ln 90 = \ln 2 + 2 \ln 3 + \ln 5$ and $\ln 150 = \ln 2 + \ln 3 + 2 \ln 5$ using exact series enclosures. Every numerical logarithm bound used here has a directed rational check in the accompanying audit.

Normalization and interpolation hypotheses

Use $n = a \cdot 2^k - 1, b_0 = 2^\ell - 1$ and $D = \ell + \beta \cdot \log_2(a) - 1$ with the preceding notation. Fix an even integer $J \geq 200$, and assume $D < J$ and $y \geq 92J$. Then $\ell \leq J, \log_2(a) < (12/7)(J+1)$, and $k > 90J$.

Let r be the unique member of $\{0,1,2,3\}$ with $k+r$ divisible by 4. Put

$$H=(k+r)/4, \alpha_1=81, \alpha_2=3^r \cdot b_0/a, \\ \Lambda=81^H - \alpha_2=3^r \cdot (a \cdot 3^k + b_0)/a > 0.$$

If $v_2(a \cdot 3^k + b_0) \leq 3$, the bounds below are immediate. Otherwise α_2 is congruent to 1 modulo 16, since 81^H is. In Bugeaud (1999), Theorem 1, rational clause (6), use $p=2$, $g=1$, $E=4$, $b_1=H$, $b_2=1$, $u=0$, and field parameters $D_{\text{field}}=e=f=1$, $t=1$. Indeed $v_2(81-1)=4>1$ and $v_2(\alpha_2-1) \geq 4 \geq 2$. Both rational arguments are 2-adic units and the linear form is nonzero. Its valuation is exactly that of $a \cdot 3^k + b_0$.

The logarithmic heights satisfy

$$h(\alpha_1)=4 \ln 3, \\ h(\alpha_2) < (12/7)(J+1) \ln 2 \leq U_4 \cdot J, \\ U_4 = (12/7) \cdot (201/200) \cdot .6932.$$

For the latter, the denominator height is already bounded. The numerator has binary logarithm less than $J+3\delta < J+5 < (12/7)(J+1)$ for $J \geq 200$. Reduction of the fraction cannot increase height. Negative rational arguments are allowed by the cited theorem.

Two explicit parameter choices

Let x and the upper limit T take the two pairs (7,120) and (9,256). For $k/J \leq T$ choose

$$M=xJ, L=5, R_1=5, S_1=1, R_2=xJ/2, S_2=10, \\ R=xJ/2+4, S=10, N=5xJ.$$

All parameters are integers because J is even, and $M \geq 1400$. Also $R_2 \leq H$ because $k > 90J$, while $x \leq 9$. The first cardinality condition holds because the five powers 81^{2z} , $0 \leq z < 5$, are distinct. The second holds because the $R_2 \cdot S_2$ values $z+Hw$ are distinct when $R_2 \leq H$, and $R_2 \cdot S_2 = ML > (M-1)L$. Congruence restrictions modulo $g=1$ are vacuous. No multiplicative-independence assumption is required here.

The factorial proof above applies, giving a normalization factor below $(23/5)/(M-1)$. Since $H \leq (TJ+3)/4$, the theorem's b obeys

$$b < (23/5) \cdot [(2x+9T)J+39]/[8(xJ-1)].$$

This ratio decreases with J . At $J=200$ it is below 90 for $(x,T)=(7,120)$, and below 150 for (9,256). Write $\gamma=1/2-N/(6RS)$, as both γ parameters of the theorem coincide when $g=1$. Exact identities imply

$$\gamma \cdot R/J = x/6 + 2/J \leq x/6 + 1/100, \\ \gamma \cdot S = 5 - 5xJ/(3xJ+24) \leq 1765/528 < 3343/1000.$$

The second expression decreases with xJ , whose minimum is 1400. Also $3 \ln N/J < .15$: with $J=200v$, $v \geq 1$, use $N \leq 9000v$, $\ln 9000 < 10$ and $\ln v \leq v-1$ to get $\ln N < 10v = J/20$.

Use the interpolation criterion with $E=4$ and $h(\alpha_1)=4 \ln 3$. Using $\ln 2 > .693$, $\ln 3 < 1.099$, $\ln 90 < 4.5$ and $\ln 150 < 5.011$ gives the strict margins after division by J :

$$7 \cdot 4 \cdot 2.772 - 7 \cdot 4.5 - .15 \\ - 5 \cdot [(353/300) \cdot 4.396 + (3343/1000) \cdot U_4] \\ = 73710029/525000000 > 0,$$

$$\begin{aligned} & 9 \cdot 4 \cdot 2.772 \cdot 9 \cdot 5.011 \cdot 15 \\ & -5 \cdot [(151/100) \cdot 4.396 + (3343/1000) \cdot U_4] \\ & = 243378343/175000000 > 0. \end{aligned}$$

The logarithm bounds have exact rational series checks. The theorem yields $v_2(\Lambda) < 4ML$. Consequently

$$\begin{aligned} & k' < 140J \text{ if } k/J \leq 120, \\ & k' < 180J \text{ if } 120 < k/J \leq 256. \end{aligned}$$

For $k/J > 256$ we switch to the following dyadic argument. Both normalizations concern the same integer $a \cdot 3^k + b_0$.

The unbounded dyadic range

Assume $D < J$ and $k/J > 256$. Put $r = k \bmod 2$, $H = (k+r)/2$, $\alpha_1 = 9$ and $\alpha_2 = -3^r \cdot b_0/a$. Then $\Lambda = 3^r \cdot (a \cdot 3^k + b_0)/a > 0$. If its valuation is at most 2 the desired bound is immediate. Otherwise α_2 is congruent to 1 modulo 8. Thus the rational criterion applies with $E=3$ and the same field and exponent parameters as before. Here $h(\alpha_1) = 2 \ln 3$ and $h(\alpha_2) < U_0 \cdot J$, where $U_0 = 525099/437500$ follows from $(12/7) \cdot (101/100) \cdot .6932$ and $J \geq 100$. Below, $\gamma > 0$.

Let q be the least integer with $k/J \leq 2^q$. Then $q \geq 9$ and $k/J > 2^{q-1}$. Choose

$$\begin{aligned} M &= (q+1)J, L = q+2, R_1 = L, S_1 = 1, \\ R_2 &= (q-1)J, S_2 = q+5. \end{aligned}$$

Thus $R = (q-1)J + q + 1, S = q + 5, N = (q+1)(q+2)J$. Here $M \geq 1000$. The inequality $q-1 < 2^{q-2}$ gives $R_2 < H$. Also $R_2 \cdot S_2 = (q^2 + 4q - 5)J \geq ML > (M-1)L$, because $q \geq 9$. Again $RS > N$ and $1/3 < \gamma < 1/2$.

The weaker factorial factor 5, and $H \leq (2^q J + 1)/2$, give

$$\begin{aligned} b &< 5 \cdot \{ [2(q-1) + (q+4)2^q]J + 3q + 4 \} / [4((q+1)J - 1)] \\ &\leq (7/4) \cdot 2^q. \end{aligned}$$

For the last step it suffices that $J[(2q-13)2^q - 10q + 10] \geq 7 \cdot 2^q + 15q + 20$. At $J \geq 100$ this follows from $(200q - 1307)2^q \geq 1015q - 980$. The latter holds for $q \geq 9$ already on replacing 2^q by 512. Hence $\ln b < .6932q + .56$.

The exact γ formulas imply

$$\begin{aligned} \gamma \cdot R/J &= q/3 - 1/6 - 2/(q+5) + (q+1)/(2J) \\ &\leq (203/600)q - 97/600, \\ \gamma \cdot S &\leq [1/2 - 350/2127]q + 5/2 - 1400/2127. \end{aligned}$$

For the second bound use $(q+1)/[(q-1)J] \leq 9/700$ and $(q+1)(q+2)/(q-1) = q + 4 + 6/(q-1)$. Finally $3 \ln N/J < q/25$: $\ln J < J/20$, $\ln(q+2) < q/3$, and $\ln(q+1) < q/3$ imply $3 \ln N/J < .15 + .02q < .04q$ at $q \geq 9$. The logarithm inequalities follow by writing $J = 100u$ or $q+2 = 11u$ and using $\ln u \leq u-1$, $\ln 100 < 4.7$ and $\ln 11 < 2.4$.

The interpolation margin, divided by J , is therefore greater than

$$\begin{aligned} P(q) &= (q+1)[(q+1) \cdot 2.079 - .6932q - .56] - q/25 \\ &\quad - (q+2) \{ [(203/600)q - 97/600] \cdot 2.198 \\ &\quad + [(1/2 - 350/2127)q + 5/2 - 1400/2127] \cdot U_0 \}. \end{aligned}$$

Its exact coefficients, from quadratic to constant, are

$$\begin{aligned} &1783170953/7444500000, \\ &-454813329/354500000, \\ &-1631430293/744450000. \end{aligned}$$

The leading coefficient is positive, $P(9)=3011630363/531750000>5.66$, and $P'(9)=1503066483/496300000>3.02$. Expansion at 9 proves $P(q)>0$ for every real $q\geq 9$. The theorem yields $k'<3J(q+1)(q+2)$. At $q=9$, $3(q+1)(q+2)=330<(38/25)\cdot 256$. The ratio of consecutive left sides is $(q+3)/(q+1)\leq 2$, so induction proves

$$3(q+1)(q+2)<(38/25)\cdot 2^{q-1}<(38/25)\cdot (k/J).$$

The first cardinality is L because $9^{2z}, 0\leq z\leq L$, are distinct. The second follows from $R_2\leq H$, which makes $z+Hw$ injective on the stated rectangle. Thus all hypotheses are checked, and this proves $k'<(38/25)k$ throughout $k/J>256$.

Retaining enough loss for the full fractional improvement

Use the elementary inequalities

$$\begin{aligned} y' &\leq \delta \cdot y - D, \\ y'' &\leq \delta \cdot y - D + \beta \cdot k'. \end{aligned}$$

If $D\geq J$, the first gives $y'\leq \delta \cdot y - J$. If $D<J$, the three ranges above give both $k'\leq \delta \cdot y - J$ and the stronger two-block endpoint

$$y''\leq \delta^2 \cdot y - (16/5)J. \quad (3)$$

For the first range use $y\geq 92J$, $\delta>19/12$ and $\beta>7/12$:

$$\begin{aligned} \delta \cdot y - k' &> (92 \cdot (19/12) - 140)J = (17/3)J, \\ \beta \cdot (\delta \cdot y - k') + D &> (119/36)J > (16/5)J. \end{aligned}$$

In the second range $k>120J$ gives $\delta \cdot y - k'>10J$ and hence a loss greater than $(35/6)J$. In the last range the bound $k'<(38/25)k$ gives a gap greater than $(19/300)k$; at $k>256J$ its product with $7/12$ exceeds $16J/5$. Each gap exceeds J , proving the intermediate run bound as well.

For $y\geq B$ choose $J=2\cdot \text{floor}(y/(2C))$, which is even and at least 200. Writing $z=y/C$, we have $J\leq z$ and $J>z-2$. Since $z\geq B/C>200$, $J\geq (99/100)y/C$. The exact inequality

$$\varepsilon \cdot C = 92/93 < 99/100$$

implies $\varepsilon \cdot y < J$. The one-block endpoint and intermediate run are therefore at most $\lambda \cdot y$. Since $\delta < 8/5$, (3) gives

$$y''\leq \delta^2 \cdot y - (16/5)J < \lambda^2 \cdot y.$$

Indeed $(\delta^2 - \lambda^2)y < 2\delta \cdot \varepsilon \cdot y < (16/5)J$. Keeping the actual two-block loss, rather than weakening it to $(\delta+1)J$, is what permits this larger ε .

Define the global extension by $G(y)=\delta \cdot y$ for $y\leq B$ and $G(y)=\max(\lambda \cdot y, y+\beta \cdot B)$ for $y\geq B$. It is continuous and increasing, satisfies $(3/2)y\leq G(y)\leq \delta \cdot y$, and equals $\lambda \cdot y$ above $2B$. Below B , apply the elementary one-block estimate. Above B , use the proved one- or two-block alternative, $G(y)\geq \lambda \cdot y$, and monotonicity. Induction along the selected restart times bounds each endpoint by the corresponding virtual iterate $G^t(y_0)$. At the single intermediate block of a two-block restart, the bound on k' gives $k_t\leq G^t(y_0)$, while $y'\leq \delta \cdot y$ gives

$y_t \leq \delta \cdot G^{t-1}(y_0)$. At endpoints the latter also follows from $G \leq \delta \cdot y$. This covers every block index.

Since $3^{28} > 2^{44}$, $(3/2)^{28} > 2B$. Every virtual orbit from $y_0 \geq 1$ reaches $2B$ by step 28. Before that point use $G \leq \delta \cdot y$, and afterwards use $G = \lambda \cdot y$, to get $G^t(y_0) \leq A \cdot y_0 \cdot \lambda^t$, including $t \leq 28$. This proves the stated all-orbit bounds.

An affine envelope and sharper cycle consequences

The same extension also satisfies, for every $y \geq 1$,

$$G(y) \leq \lambda \cdot y + \varepsilon \cdot B.$$

For $y \leq B$ this is $\varepsilon \cdot y \leq \varepsilon \cdot B$. For $y \geq B$ the linear term is immediate; for the other term subtracting the right side leaves $(\lambda-1)(B-y) \leq 0$. Set $c = \varepsilon \cdot B / (\lambda-1)$. Induction gives

$$G^t(y_0) \leq (y_0 + c) \cdot \lambda^t - c. \quad (4)$$

For a hypothetical nontrivial cycle, rotate to its minimum. Simons-de Weger [2], Theorem 3(d), gives $K < 16m \cdot \delta^m$ for $m \geq 91$ after coarsening its three applicable coefficients. Corollary 13 gives $n_{\min} < m \cdot \exp(13.3 \cdot (.46057 + \ln K))$. Consequently, with $a_0 = \log_2(\delta)$,

$$y_0 < 1 + 13.3 \cdot .46057 / \ln 2 + 53.2 + 14.3 \log_2(m) + 13.3 a_0 \cdot m.$$

For $m \geq 1024$ use $\ln 2 > .69$, $a_0 < 2/3$, and $\log_2(m) \leq m/64$. The constant term is below 64, and $13.3 \cdot (2/3) + 14.3/64 + 64/1024 < 10$. The logarithm inequality holds at 1024 and its difference increases thereafter. The bound $a_0 < 2/3$ follows from $\delta < 317/200$ and $(317/200)^3 < 4$. Thus $y_0 < 10m$. Exact rational coefficient bounds now imply

$$K < 21.08m \cdot \lambda^m, \quad m \geq 1024,$$

$$K < 1.51m \cdot \lambda^m, \quad 1024 \leq m \leq 515619.$$

For the second bound, Theorem 3(d) gives $n_{\min} < 51825000 \cdot m^2 \cdot \delta^m$ when $91 \leq m \leq 515619$. Hence

$$y_0 < Y(m) = 1 + \log_2(51825000) + 2 \log_2(m) + a_0 \cdot m.$$

Since $51825000 < 2^{26}$ and $\log_2(m)/m$ decreases for $m \geq 1024$, $Y(m)/m < 2/3 + 47/1024$. The exact audit checks $[A/(\lambda-1)] \cdot [2/3 + 47/1024] < 1.51$. Summing the block estimates therefore gives the stated middle-range coefficient.

There is a stronger large-range coefficient. Substitute the first bound into Simons-de Weger, Corollary 13. With

$$Q = 1 + 13.3 \cdot .46057 / \ln 2 + 13.3 \log_2(21.08),$$

we obtain $y_0 < Q + 14.3 \log_2(m) + 13.3m \cdot \log_2(\lambda)$. Summing (4) around the cycle consequently gives

$$\begin{aligned} K &< [(y_0 + c)/(\lambda-1)] \cdot \lambda^m, \\ K &< 15.17m \cdot \lambda^m, \quad m \geq 515620. \end{aligned} \quad (5)$$

For (5), the coefficient upper bound

$$[13.3 \log_2(\lambda) + (Q + 14.3 \log_2(m) + c)/m] / (\lambda-1)$$

decreases for $m \geq 515620$, since both $1/m$ and $\log(m)/m$ decrease there. At $m = 515620$ an exact upward rational calculation puts it below 15.17. This uses an already established bound as input to Corollary 13, so the argument is not circular. The affine estimate improves the coefficient; the fourth-power normalization and retained loss improve the base.

Verification and prior work

The written proof covers all parameter values; finite numerical challenges only supplement it. Exact rational audits check the constants, and Lean checks selected algebra, congruence, local-loss, envelope and restart lemmas. The published p-adic theorem, the full specialization, and the cycle-size inputs are not formally verified here. A domain expert should review the source hypotheses and both universal parameter arguments.

The normalized-exponent method is established transcendence theory [3]. Luca [4] proves a related effective relation between cycle complexity and height, using a run parameter that counts blocks with $k_i \geq 2$ and can be smaller than m . The explicit smaller base in this manuscript still needs a complete priority review. None of these estimates excludes arbitrary cycles or proves the Collatz conjecture.

Primary references

- [1] Y. Bugeaud, Linear forms in p-adic logarithms and the Diophantine equation $(x^n-1)/(x-1)=y^q$, Mathematical Proceedings of the Cambridge Philosophical Society 127(3)(1999),373-381, Theorem 1, clause (6). DOI [10.1017/S0305004199003692](https://doi.org/10.1017/S0305004199003692) Author's complete PostScript, pages 3-4 visually inspected after local rendering: [Author-hosted complete PostScript](#) The author version explicitly uses logarithmic heights. A comparison with the final journal text remains desirable; no unread text is assumed.
- [2] J. Simons and B. de Weger, Theoretical and computational bounds for m-cycles of the $3n+1$ problem, version 1.44, 31 August 2010, Theorem 3(d) and Corollary 13. [Author-hosted version 1.44](#)
- [3] Y. Bugeaud, B-prime, arXiv:2209.00275(2022), Section 1, for the history of normalized-exponent refinements: [arXiv:2209.00275](https://arxiv.org/abs/2209.00275)
- [4] F. Luca, On the non-trivial cycles in Collatz's problem, SUT Journal of Mathematics 41(1)(2005),31-41. Theorem 2.1 is relevant prior work on exponential cycle-complexity bounds. Its run parameter counts the blocks with $k_i \geq 2$, and can be smaller than m . Its displayed height recurrence uses a geometric factor at least δ , rather than the explicit smaller base established in the present derivation. [Publisher PDF](#)